TEMASEK POLYTECHNIC
SCHOOL OF INFORMATICS & IT
DIPLOMA IN CYBERSECURITY & DIGITAL FORENSICS
AY2023/2024 OCTOBER SEMESTER
SECURITY TECHNOLOGY AND INNOVATION
PROBLEM PACKAGE SPECIFICATIONS (100%)

SECURITY TECHNOLOGY AND INNOVATION (CCD3C01)
SUBJECT LEVEL: 3

TIME ALLOWED: Refer to teaching plan

INSTRUCTIONS TO CANDIDATES

1. This problem package specification consists of 5 pages (including cover page).

2. This problem package specification contributes 100% of the subject.

3. Breakdown of mark weightage as follows: -
    a. Part I - Research and Findings (30%)
        i. 20% Individual effort
        ii. 10% Group effort
    b. Part II – Proposal and Documentation
        i. 40% Individual effort
    c. Part III – Presentation
        i. 30% Individual effort

4. This problem package is to be done in group of 4.

5. All written reports per group MUST be submitted via PoliteMall (safe-assign is enabled). Refer to the teaching plan for the deadline.

   Your zipped document file should be named according to the following format:

   YourName_StudentID_ YourClass.zip

   e.g.: Teo_Chu_Chu_1234567D_P01.zip

   <u>Note: Do NOT compress in RAR format.</u>

6. The presentation shall be held during your lab sessions. Refer to your time table for the schedule.

Penalty for Late Submission

late and < 1 day	: 10% deduction from absolute mark given for the project
late >= 1 and < 2 days	: 20% deduction from absolute mark
late >= 2 days	: No marks awarded

Note that "day" includes **non-working days** (Sat, Sun and public holidays).

General MC/LOA is NOT considered as valid reason for extended project submission.

Penalty for Plagiarism

Once discovered, plagiarised work and the student involved will be reported to the school authority and zero mark will be given for the project. Please refer to ChatGPT Playbook Student Edition for ChatGPT usage in this project.

ChatGPT
Playbook_Student Edi

Please include attached Declaration of Originality in your submission.

Template -
Declaration of Origina

Introduction

This subject aims to equip you with the knowledge and skills to research on the latest trends, conceptualize a problem statement and construct innovative proof-of-concept for your IT security solution which is based on tackling emerging security trends and demands.

Project Scope

In this project you will work as a team to propose and create a prototype based on the problem package specifications defined in this document. Your project should focus on cybersecurity use cases, e.g. Compliance, Forensics or Penetration Testing, etc.

Your solution will be evaluated based on innovative, creativity, originality and practicality. You will also be judged on the completeness of your prototype.

Project Deliverables

This assessment contributes 100% of the subject. Your deliverables will include Research and Findings, Proposal and Documentation and finally, a Presentation.

<u>Research and Findings (10% Group & 20% Individual)</u>
This component consists of 2 parts: -
1. Session 1-4 PPT Slides (5% Group)
2. Research and Findings (5% Group & 20% Individual)

For the 1st part - Session 1-4 PPT slides components, you will be required to work together with your project group members to research and create PPT slides on the followings topics: -
1. Self-Directed Learning (SDL)
2. Scrum framework
3. Emerging cybersecurity trends
4. Problem statement and FILA chart
5. SDL – Plan, Perform, Monitor and Reflect (PPMR)

For the 2nd part - Research and Findings component, you will be required to provide a report with the followings: -
1. Research on emerging security trends
2. Pick one of the trends and brainstorm for ideas for your project.
    a. Create a problem statement and propose a solution to address it.
    b. **Create multiple mini problem statements for each of your group members to work on. There should be at least 1 mini problem statement per group members.**
    c. Provide a FILA chart on these problem statements.*
3. Explain the relevance of your solution to cybersecurity.

4. Provide drawings or drafts of your invention prototype.
5. Describe the novelty or originality of your invention.
6. Provide a comparison chart of your invention as compared to existing or alternative products. Explain what sets your invention apart from previous creations.
7. List down the practical applications and usefulness of the invention.
8. Provide supporting materials that assist the evaluator in evaluating your ideas. eg. trials, papers, literature or correspondence.
9. Provide a project schedule.*

* These items are assessed as part of Group efforts and the rest are assessed as individual.

Session 1-4 PPT Slides and Research and Findings Submission Details

Deliverable	Details	Weightage
Session 1-4 PPT slides deliverables	Submission of PPT slides by the end of each session.	5% Group
Research and Findings	Submission of Fila chart and project plan. Refer to Teaching Plan for the date of submission.	5% Group 20% Individual

Proposal and Documentation (40% Individual)

The proposal and documentation component will be used to assess students' ability to develop the proposal and related documentation as well as a prototype of the solution.

Complete the remaining sections of your report in terms of

- Introduction
- Relevance of invention to cybersecurity
- Work distribution
- Data classification diagram
- Use and misuse case diagram
- Experience as an inventor
- Commercial Feasibility
- Potential Market Opportunity
- Conclusion
- Reflection

Proposal and Documentation Submission Details (40%)

Deliverable	Details	Weightage
2nd deliverable	Submission of Interim report. See P0xGpx_NAME_STI_FinalReport_template.doc for the template. Refer to Teaching Plan for the date of submission.	40% Individual

Presentation (30% Individual)

The completeness of the prototype and presentation will be used to assess students' performance in this component. Prototype will be assessed on functionality, user friendliness and completeness of the product. Presentation will be assessed based on ability to articulate in a professional manner as if presenting to a group of potential investors.

Deliverable	Details	Weightage
3rd deliverable	Formal Presentation and Demonstration of Final Product as a group. Refer to Teaching Plan for the date of presentation.	30% Individual

Breakdown of assessment weightages is as follows.

Assessment	Weightages	
	Individual	Group
• Research and Findings • Proposal and Documentation • Presentation	20% 40% 30%	10%
Total	**90%**	**10%**

You will be assessed on the following:

- Creativity of the invention
- Originality of the invention
- Practicality of the invention
- Completeness of the prototype
- Documentation
- Presentation
- Individual questions and answers

Please also take note of the standard penalties for late submission (refer to teaching plan), which will apply for this project. Also remind yourself of the heavy penalty for plagiarism.

Problem Package Specification:

What?
You are to create a prototype to tackle issues resulting from emerging cybersecurity trends or leverage on emerging technologies used in the field of cybersecurity.

Problem Statement
Based on the “what?” that your project will be tackling, define your problem statement. Please provide details of your project’s objectives. Prepare a detailed proposal which must cover the following requirements:

- Project must solve a real world issue
- Either tackle issues resulting from emerging cybersecurity trends or leverage on emerging technologies used in the field of cybersecurity

Project Deliverables
To pass this subject, you will need to submit a prototype of the project which is able to demonstrate your solution’s features.

THE END